

Executive Data Analysis Report

Project Identifier: AI-DATA-ANALYST-20260725-223826

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1. Executive Summary

The model achieves an OVERALL ACCURACY of 74.3% when predicting late delivery on orders. The ROC-AUC of 0.769 indicates the model's ability to rank orders that resulted in late delivery higher than orders that did not result in late delivery. At a standard threshold, the model correctly identified 1,021 orders with late delivery. However, it missed 544 orders that actually experienced cases of late delivery. Furthermore, it raised false alarms for 4,416 orders that did not have late delivery. The model correctly left 13,315 orders without a late delivery alert.

2. Data Diagnostics Profile

Input shape	96,476 rows × 23 columns
Output shape	96,476 rows × 22 columns
Duplicate rows removed	0
High-missing columns flagged	None
Low-variance columns dropped	order_status
Columns type-corrected	None
Numeric columns imputed	customer_zip_code_prefix, n_items, n_distinct_sellers, n_distinct_products, total_price, total_freight, avg_product_weight_g, avg_product_photos_qty, total_payment_value, max_installments, n_payment_methods, purchase_to_estimated_days, is_late_delivery
Categorical columns imputed	order_id, order_status, order_purchase_timestamp, order_estimated_delivery_date, customer_unique_id, customer_city, customer_state, primary_seller_state, product_category, primary_payment_type

3. Model Performance Leaderboard

Task type: binary_classification | **Best model:** HistGradientBoostingClassifier

Candidate Model	CV Score
HistGradientBoostingClassifier	0.5721
RandomForestClassifier	0.5660
LogisticRegression	0.4749

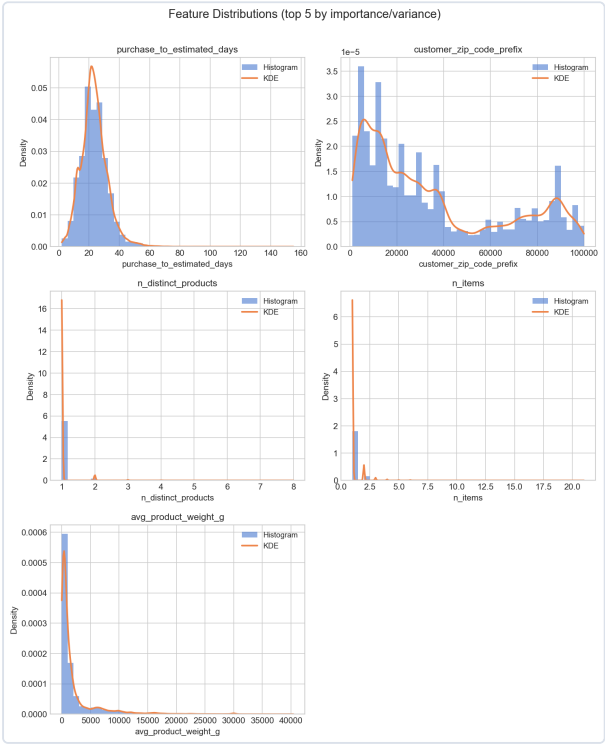
F1 (macro)	0.5673
Precision (macro)	0.5743
Recall (macro)	0.7017
ROC-AUC	0.7693

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	13,315	4,416
Actual Positive (1)	544	1,021

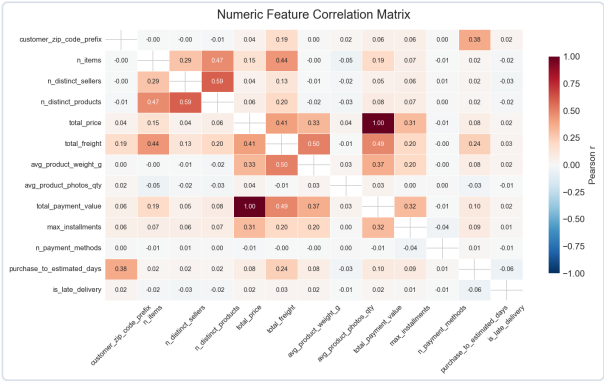
Threshold	Precision	Recall	F1 (macro)
0.3	0.114	0.897	0.379
0.4	0.149	0.802	0.494
0.5	0.188	0.652	0.567

4. Automated Business Insights

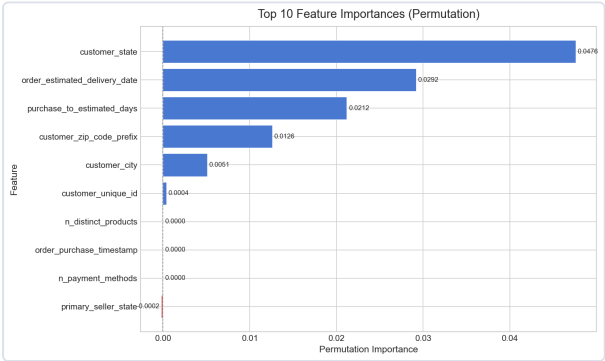
Customer state matters because different states have unique logistics routes and infrastructure, making some orders from specific states more prone to cases of late delivery. Order estimated delivery date matters because orders scheduled during peak demand periods face tighter shipping windows, raising the likelihood of late delivery on those orders. Purchase to estimated days matters because a longer waiting period between purchase and planned arrival gives carriers more opportunities for delays, creating cases of late delivery in those orders. Customer zip code prefix matters because specific local neighborhoods have varying last-mile challenges, causing certain orders to experience late delivery more frequently.



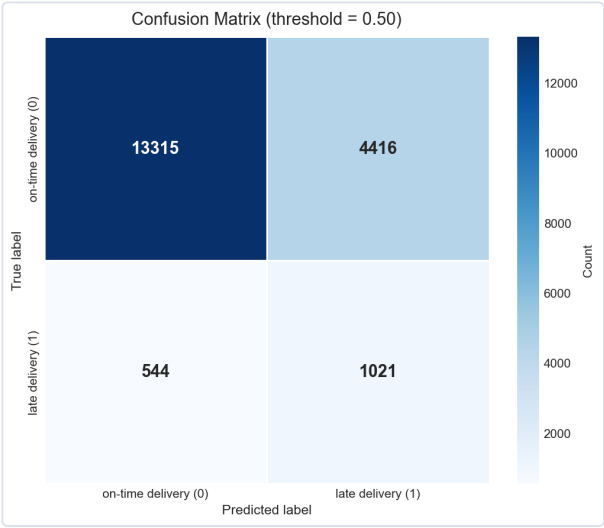
Distributions



Correlation Heatmap



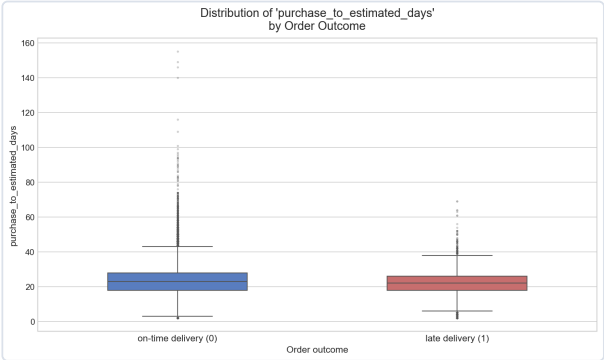
Feature Importance



Confusion Matrix



Threshold Tradeoff



Top Feature Vs Target

5. Strategic Recommendations

To maximize catching cases of late delivery, use threshold=0.3; at this tradeoff, the precision is 0.114 and the recall is 0.897. To minimize unnecessary customer communications on orders that will arrive on time, use threshold=0.5; at this tradeoff, the precision is 0.188 and the recall is 0.652. To balance catching a high volume of late orders without overwhelming operations, use threshold=0.4; at this tradeoff, the precision is 0.149 and the recall is 0.802.